

Exp No:1 (b)**Implementation of Lexical analyzer using LEX****Aim:**

To implement a Lexical Analyzer using LEX to identify keywords, identifiers, numbers, operators, and special symbols from the given input.

Algorithm:

1. Start the program.
2. Create a LEX specification file with the extension .l.
3. Define regular expressions for keywords, identifiers, numbers, operators, and special symbols.
4. Define actions to display the type of each recognized token.
5. Save the LEX program as lexer.l.
6. Open Command Prompt and navigate to the folder containing lexer.l.
7. Execute flex lexer.l to generate the C source file lex.yy.c.
8. Compile the generated C file using gcc lex.yy.c -o lexer.exe.
9. Execute the generated program using lexer.exe.
10. Enter the input statement.
11. The lexical analyzer identifies and displays each token.
12. Stop the program.

Program:

```
%{
#include <stdio.h>
%}

%%

"int"|"float"|"char"|"if"|"else"|"for"|"while"|"return"
    { printf("%s -> Keyword\n", yytext); }

[a-zA-Z_][a-zA-Z0-9_]*
    { printf("%s -> Identifier\n", yytext); }

[0-9]+
    { printf("%s -> Number\n", yytext); }

"+"|"-"|"*"|"/"|"="|"<"|">"
    { printf("%s -> Operator\n", yytext); }

"("|")"|"{"|"}"|";"|"|"
    { printf("%s -> Special Symbol\n", yytext); }

[ \t\n]+
    { }

.
    { printf("%s -> Unknown Symbol\n", yytext); }
```

```
%%
```

```
int yywrap()  
{  
    return 1;  
}
```

```
int main()  
{  
    printf("Enter the input:\n");  
    yylex();  
    return 0;  
}
```

Procedure to Execute:

Step 1: Create the LEX file

Open **Notepad** and paste the above LEX program.

Save the file as:

lexer.l

While saving, select **Save as type** → **All Files**.

Make sure the file is saved as lexer.l and **not** lexer.l.txt.

Step 2: Open Command Prompt

Open CMD and move to the folder where lexer.l is saved.

For example:

cd Desktop

Check the file:

ls

The file lexer.l should be visible.

Step 3: Generate the C source file

Type:

flex lexer.l

If there is no error, a file named:

lex.yy.c

will be generated.

Step 4: Compile the generated C file

Type:

gcc lex.yy.c -o lexer.exe

If there is no error, the executable file:

lexer.exe

will be created.

Step 5: Run the Lexical Analyzer

Type:

lexer.exe

The program will display:

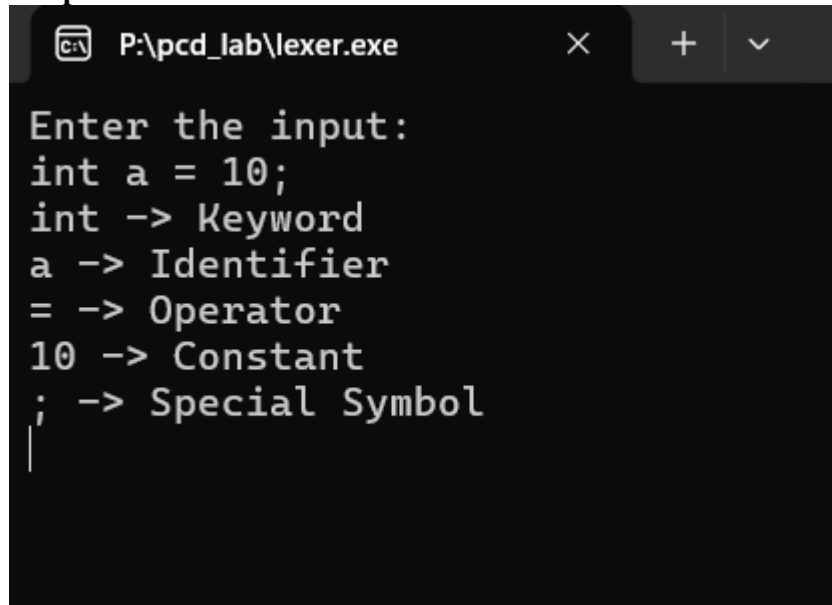
Enter the input:

Enter the following input:

int sum = a + 25;

Press **Enter**.

Output:

A screenshot of a Windows command prompt window titled "P:\pcd_lab\lexer.exe". The window shows the output of the lexical analyzer for the input "int a = 10;". The output is as follows:

```
Enter the input:
int a = 10;
int -> Keyword
a -> Identifier
= -> Operator
10 -> Constant
; -> Special Symbol
|
```

Result:

Thus, the Lexical Analyzer was successfully implemented using LEX, and the given input was successfully analyzed into keywords, identifiers, numbers, operators, and special symbols.